CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 8 pipes fill a tank in 9 hours. 12 such pipes will fill it in: 1

(a) 13.5 hours **(b)** 5 hours **(c)** 6 hours **(d)** 7 hours

2. A : B = 1 : 2 and B : C = 4 : 3. If A + B + C = 135, then C equals: 1

(a) 30 **(b)** 45 **(c)** 27 **(d)** 60

3. On a map with scale 1 : 200000, a distance of 3 cm represents: 1

(a) 600 km **(b)** 9 km **(c)** 60 km **(d)** 6 km

4. 54 km/h expressed in m/s is: 1

(a) 18 m/s **(b)** 194.4 m/s **(c)** 15 m/s **(d)** 0.9 m/s

5. Two numbers are in the ratio 3 : 10 and their sum is 91. The larger number is: 1

(a) 70 **(b)** 21 **(c)** 77 **(d)** 45

6. Two numbers are in the ratio 4 : 9. If 5 is added to each, the numbers become 25 and 50. The original smaller number is: 1

(a) 15 **(b)** 25 **(c)** 20 **(d)** 40

7. A mixture of 90 L has milk and water in the ratio 4 : 5. The quantity of water is: 1

(a) 45 L **(b)** 55 L **(c)** 40 L **(d)** 50 L

8. 5 pipes fill a tank in 12 hours. 15 such pipes will fill it in: 1

(a) 36 hours **(b)** 5 hours **(c)** 4 hours **(d)** 2 hours

9. A : B = 3 : 4 and B : C = 8 : 7. If A + B + C = 210, then C equals: 1

(a) 42 **(b)** 60 **(c)** 80 **(d)** 70

10. On a map with scale 1 : 100000, a distance of 7 cm represents: 1

(a) 14 km **(b)** 70 km **(c)** 700 km **(d)** 7 km

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Convert 63 km/h into m/s. 2

12. A map has scale 1 : 200000. Two places are 4 cm apart on the map. Find the actual distance in km. 2

13. Two numbers are in the ratio 7 : 8 and their sum is 75. Find the numbers. 2

14. Convert 81 km/h into m/s. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A mixture of 45 L contains milk and water in the ratio 4 : 5. How much water must be added to make the ratio 4 : 8? 3

16. A : B = 2 : 3 and B : C = 9 : 4. If A + B + C = 255, find the three numbers. 3

17. The ages of a mother and daughter are in the ratio 9 : 4. The mother is 45 years old. Find their ages 5 years ago and the ratio then. 3

OR

A mixture of 72 L contains milk and water in the ratio 5 : 3. How much water must be added to make the ratio 5 : 6? 3

18. A : B = 1 : 3 and B : C = 9 : 7. If A + B + C = 195, find the three numbers. 3

OR

The ages of a mother and daughter are in the ratio 8 : 3. The mother is 32 years old. Find their ages 2 years ago and the ratio then. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. 9 men can complete a piece of work in 40 days. 5
(i) How many days will 15 men take?
(ii) How many men are needed to finish it in 12 days?

OR

Anil and Bhanu start a business with ₹ 20000 and ₹ 30000. They share a profit of ₹ 2000 in the ratio of their investments. Find the ratio of investments and each person's share. 5

20. 14 men can complete a piece of work in 18 days. 5
(i) How many days will 21 men take?
(ii) How many men are needed to finish it in 12 days?

OR

Anil and Bhanu start a business with ₹ 14000 and ₹ 21000. They share a profit of ₹ 1750 in the ratio of their investments. Find the ratio of investments and each person's share. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND